

A null result for the matrix-valued sharp one-dimensional Lieb–Thirring inequality, and a Comparator check of the scalar Lean formalisation

Xavier Callens (SocrateAI Lab)

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Abstract

We report two things, neither of which is a mathematical breakthrough. First, an adversarial numerical search for a violation of the matrix-valued sharp one-dimensional Lieb–Thirring inequality $\text{Tr}(-\partial_x^2 \otimes 1_m - W)_-^\gamma \leq L_1(\gamma) \int \text{tr} W^{\gamma+1/2} dx$ for $1/2 < \gamma < 3/2$, with L_1 the one-bound-state constant. The instrument is one-sided by construction (the computed numerator is a lower bound of the true one; the ratio is a lower bound up to quadrature and truncation error, which we measured at the 10^{-7} – 10^{-6} level), and was validated by positive and negative controls before any result was read. Across 22 cells at $m = 2$ (336 restarts), 6 cells at $m = 3$, a second-variation test at the embedded scalar extremiser, and an independent search of the dual kinetic form, no violation was found: 0 restarts exceeded L_1 by the flag threshold, and the closest approach is 2.84×10^{-7} below L_1 , which is below the instrument’s resolution of about 10^{-6} . This does *not* prove the matrix inequality. Second, using upstream’s pinned Lean and Mathlib (the Comparator tools themselves built from their v4.34.0 tags), we ran `leanprover/comparator` on the scalar sharp Lieb–Thirring theorem claimed in the `openai/math` repository; the Lean default kernel accepted it with only the three standard axioms (no external kernel was run). The Lean statement is scalar; the Comparator certifies that the solution proves the challenge statement, not that the statement is a faithful formalisation. If it is faithful, the theorem would settle the scalar sharp inequality for $1/2 < \gamma < 3/2$, which we understand to be an open conjecture (we did not search the literature exhaustively); that is why the statement audit, which has not been done, matters.

1 Claim, scope and what is *not* claimed

For $m \geq 1$, a matrix potential $W \geq 0$ on $\mathbb{R}$ with values in $m \times m$ Hermitian matrices, and $\gamma \in (1/2, 3/2)$ the inequality in question is

$$\text{Tr}(-\partial_x^2 \otimes 1_m - W)_-^\gamma \leq L_1(\gamma) \int_{\mathbb{R}} \text{tr} W(x)^{\gamma+1/2} dx, \quad L_1(\gamma) = \frac{2\left(\frac{\gamma-1/2}{\gamma+1/2}\right)^{\gamma-1/2} \Gamma(\gamma+1)}{2\sqrt{\pi} \Gamma(\gamma + \frac{3}{2})}. \quad (1)$$

Here L_1 is the constant for which equality holds at the one-bound-state potential $W = (r + 1) \text{sech}^2(rx) ee^\dagger$, e a unit vector, with $r = 1/(\gamma - \frac{1}{2})$ (one bound state $\text{sech}^{\gamma-1/2}(rx) e$ at energy -1). For $m = 1$ this is the scalar case of the Lieb–Thirring conjecture [2] with the one-bound-state constant; for $\gamma \geq 3/2$ the operator-valued inequality with the *classical* constant is a theorem of Laptev and Weidl [1]. The classical constant coincides with L_1 only at $\gamma = 3/2$ (where it equals $3/16$), which we use as a control. The statement that (1) holds for $m \geq 2$ in the range $1/2 < \gamma < 3/2$,

together with a classification of the equality cases, appears as a headline claim of one family in the public `openai/math` repository (accessed 2026-10-08), which presents it as the work of an AI model. We did not assess its literature status. The matrix claim is *not* part of the Lean formalisation in that repository, which is scalar.

What this paper does not claim. It does not prove or refute (1) for $m \geq 2$; it does not show the Lean statement is an adequate formalisation of any conjecture; it contains no new theorem. A search that finds nothing is evidence of the instrument’s reach, not of truth.

2 Instrument and preregistration

One-sided design. The eigenvalues of $-\partial_x^2 \otimes 1_m - W$ are computed on the Dirichlet box $[-L, L]$ ($L = 24$) in a sine basis by Rayleigh–Ritz. Dirichlet restriction can only raise eigenvalues (domain monotonicity) and a finite basis raises them again (min–max), so every computed negative eigenvalue lies above the true one, and the computed numerator $\sum |E_j|^\gamma$ is a lower bound of the whole-line numerator, up to quadrature error in the matrix elements (Gauss–Legendre, $Q = 1200$ nodes; $L = 24$ in the campaigns, $L = 14$, $Q = 1600$ in the Hessian runs). The denominator $\int \text{tr } W^{\gamma+1/2}$ is also computed by quadrature on the box, so the ratio is a lower bound only up to quadrature and truncation error, which we measured at the 10^{-7} to 10^{-6} level. Potentials are $W = B^\dagger B$ with B built from sech times Chebyshev polynomials in $\tanh$ with complex coefficients and shifts. Candidate violations were to be re-evaluated on a refined grid and by an independent finite-difference scheme with Richardson extrapolation (agreement about 10^{-6}); a candidate counted as certified only if the refined-grid excess exceeded 10^{-5} and the finite-difference excess exceeded 5×10^{-6} . The verdict rule was interval-style and abstaining: VOID, VIOLATION_CERTIFIED, UNDECIDED, NO_VIOLATION_FOUND.

Preregistration. Hypotheses H-LT1 (no violation at $m = 2$, then $m = 3$), H-LT2 (structure of near-maximisers), H-LT3 (second variation at the embedded scalar extremiser), H-LT4 (instrument validity) and H-LT5 (dual kinetic form) were registered with kill criteria and the sha256 of the frozen instrument and campaign code before any campaign cell was run; a few smoke values had been seen before registration (listed in the preregistration file), the control rule C2 of the dual-form lane was relaxed after a first failure (disclosed in its preregistration), and registration times are local-clock commits. Seeds were ledger-enforced and never reused. Deviations are logged in each lane’s `deviations.md`.

Controls. Table 1 lists the controls, run before the campaigns. The two negative controls matter: at $\gamma = 3$ the one-soliton constant is not optimal, and the same code must find a violation; it does. The exact extremiser itself is computed *below* L_1 by up to 6.5×10^{-7} (C1). The instrument therefore cannot resolve differences below about 10^{-6} .

3 Results

3.1 $m = 2$ (H-LT1)

22 cells over $\gamma \in \{0.75, 1, 1.25, 1.4\}$ and the families random, embedded-scalar, rotated-pair and x -dependent twist, with $K \in \{2, 3\}$ and $P \in \{8, 10\}$, 336 restarts in total. Table 2 gives, per cell, how far below L_1 the best restart stays. No restart reached the flag threshold of 10^{-7} above L_1 .

id	test	measured	result
C1	equality potential, $\gamma \in \{0.75, 1, 1.25, 1.4\}$	computed ratio minus L_1 in $[-6.48 \times 10^{-7}, -2.36 \times 10^{-11}]$, one bound state	pass
C2/C3	unitary invariance, embedding, additivity ($m = 2, 3$)	residuals $\leq 1.08 \times 10^{-15}$	pass
C4	scalar search ($m = 1$) must reach but not exceed L_1	best excess $\leq -2.21 \times 10^{-6}$	pass
C6	$\gamma = 3/2$, proved sup 3/16, matrix search	best excess -2.55×10^{-7} (not exceeded)	pass
C5	NEGATIVE: wrong claimed constant at $\gamma = 3$	violation found, excess 0.1595	pass
C7	NEGATIVE: matrix search at $\gamma = 3$	violation found, excess 0.1595	pass

Table 1: Controls (from `results/openai_math/hypotheses/lt_matrix/controls.json`). Excess means computed ratio over L_1 minus one.

The closest approach, 2.84×10^{-7} below L_1 , is below the resolution of about 10^{-6} (at $\gamma = 1$ the exact extremiser itself is computed 6.5×10^{-7} below L_1): the search found nothing better than the scalar soliton, and could not rule out a violation smaller than about 10^{-6} . The cells whose best restart is farthest below L_1 stay 2.15×10^{-5} below it.

3.2 $m = 3$ (H-LT1)

Table 3 lists the six cells and the $\gamma = 3/2$ control, where the supremum is a proved theorem and must not be exceeded; it was not.

3.3 Second variation at the embedded scalar extremiser (H-LT3)

Perturbing $B^* = \text{diag}(\sqrt{r+1} \text{sech}(rx), 0, \dots, 0)$ within a single-blob, degree-8 parametrisation (through a first-order-eigenvector surrogate of the Hessian), excluding the exactly flat directions (translation, dilation), the restricted Hessian of $\log R$ has 0 positive eigenvalues beyond a relative tolerance of 10^{-6} times the spectral norm in all 12 rows of Table 4. These are only four independent values of γ : the largest restricted eigenvalue is the same for $m = 1, 2, 3$ by construction. The same code on the $\gamma = 3$ negative control returns confirmed positive directions (Table 5; 36 positive eigenvalues, all in the control); that control is evaluated at a non-critical point, so it shows only that positive curvature is reported when present. The correct wording is “no positive direction found beyond tolerance”: most modes are flat to numerical precision (122 of 162 at $m = 3$), so this is *not* a proof that the embedded extremiser is a strict local maximum.

3.4 An independent search space: the dual kinetic form (H-LT5)

At $\gamma = 1$, (1) implies for orthonormal families $\psi_1, \dots, \psi_N$ in $L^2(\mathbb{R}; \mathbb{C}^m)$, with $\rho = \sum_j \psi_j \psi_j^\dagger$, that $\sum_j \|\psi_j'\|^2 \geq \frac{\pi^2}{4} \int \text{tr} \rho^3 dx$, because $4/(27L_1(1)^2) = \pi^2/4$. The kinetic energy is exact in the sine basis and $\int \text{tr} \rho^3$ is evaluated on a periodic grid and re-checked on a 2^{24} grid and by Gauss–Legendre. The implication goes one way only: a violation of the dual form would violate (1), but not conversely,

γ	family	K/P	restarts	best distance below L_1	flagged
0.75	embed	2/8	12	8.95×10^{-7}	0
0.75	random	2/8	12	1.72×10^{-6}	0
0.75	random	3/10	12	9.70×10^{-6}	0
0.75	rotpair	2/8	12	3.85×10^{-6}	0
0.75	twist	2/8	24	6.04×10^{-7}	0
1.0	embed	2/8	12	9.55×10^{-7}	0
1.0	random	2/8	12	2.66×10^{-6}	0
1.0	random	3/10	12	9.56×10^{-6}	0
1.0	rotpair	2/8	12	2.43×10^{-6}	0
1.0	twist	2/8	24	6.21×10^{-7}	0
1.25	embed	2/8	12	6.40×10^{-7}	0
1.25	random	2/8	12	4.73×10^{-6}	0
1.25	random	3/10	12	2.15×10^{-5}	0
1.25	rotpair	2/8	12	1.30×10^{-6}	0
1.25	twist	2/8	36	2.84×10^{-7}	0
1.25	twist	3/10	12	9.16×10^{-7}	0
1.4	embed	2/8	12	9.52×10^{-7}	0
1.4	random	2/8	12	4.77×10^{-6}	0
1.4	random	3/10	12	1.39×10^{-5}	0
1.4	rotpair	2/8	12	1.33×10^{-6}	0
1.4	twist	2/8	36	3.41×10^{-7}	0
1.4	twist	3/10	12	8.31×10^{-7}	0

Table 2: All $m = 2$ cells (from `LT_A/result.json`). Restarts per cell are unequal (12, 24 or 36). The “twist” cells received extra restart chunks; the preregistered saturation criterion was not met for $\gamma = 0.75$ and 1.

so this null result is weaker evidence than the spectral search. Table 6 gives, for $m \in \{1, 2, 3\}$ and $N \in \{1, 2, 3, 4\}$, the smallest relative excess of the quotient over $\pi^2/4$ found in 12 restarts per cell; the smallest value over all 12 cells is 1.49×10^{-8} above, so nothing went below. In 3 cells (those with $(m, N) \in \{(1, 3), (1, 4), (2, 4)\}$) no restart came within 10^{-5} of $\pi^2/4$ although for $m = 1$ the quotient is bounded below by $\pi^2/4$ if upstream’s scalar theorem is correct and its Lean statement faithful (we audited neither); the search resolution there is poor, and a violation smaller than about 10^{-4} would be invisible.

3.5 Structure of near-maximisers (H-LT2): the preregistered prediction was killed

The preregistered prediction was that near-maximisers are, up to a constant unitary, direct sums of scalar solitons: a commutator statistic $C < 0.01$. On a snapshot of the data taken during the night, 345 near-maximiser rows existed, of which 125 had stored parameters (220 did not and were not analysed). Of these, 59 met the kill rule ($R \geq (1 - 10^{-4})L_1$ with $C > 0.05$), 52 of them outside the $\gamma = 3/2$ control; 47 were intermediate and 19 held the prediction. An exploratory, not preregistered, follow-up suggests that kills from embedded and twisted starts are mostly tail noise of a scalar soliton plus a zero channel (one rotated-pair row at $\gamma = 1.4$ is a non-commuting single-channel exception); about half of the random-family kills remain non-commuting at the stricter threshold and are multi-channel, and for the rest tail noise is not excluded. Near-maximisers here means within the instrument resolution of L_1 . We offer an untested explanation: spatially separated pieces with different eigenvector directions decouple asymptotically (so $R \approx L_1$) without commuting

cell	restarts	best distance below the bound
control, $\gamma = 3/2$	8	2.02×10^{-7}
$\gamma = 1.0$, embed	8	2.75×10^{-6}
$\gamma = 1.0$, random	8	1.38×10^{-5}
$\gamma = 1.0$, twist	8	8.23×10^{-7}
$\gamma = 1.25$, embed	8	1.10×10^{-6}
$\gamma = 1.25$, random	8	9.25×10^{-6}
$\gamma = 1.25$, twist	8	8.99×10^{-7}

Table 3: $m = 3$ cells (from `LT_B/result.json`).

γ	m	$\lambda_{\max}$ (restricted)	tolerance	>tol	within tol	< -tol
0.75	1	2.93×10^{-9}	6.84×10^{-7}	0	10	8
0.75	2	2.93×10^{-9}	6.84×10^{-7}	0	48	24
0.75	3	2.93×10^{-9}	6.84×10^{-7}	0	122	40
1.0	1	3.46×10^{-9}	1.11×10^{-6}	0	10	8
1.0	2	3.46×10^{-9}	1.11×10^{-6}	0	48	24
1.0	3	3.46×10^{-9}	1.11×10^{-6}	0	122	40
1.25	1	3.12×10^{-10}	1.85×10^{-6}	0	10	8
1.25	2	3.12×10^{-10}	1.85×10^{-6}	0	48	24
1.25	3	3.12×10^{-10}	1.85×10^{-6}	0	122	40
1.4	1	1.82×10^{-10}	2.66×10^{-6}	0	10	8
1.4	2	1.82×10^{-10}	2.66×10^{-6}	0	48	24
1.4	3	1.82×10^{-10}	2.66×10^{-6}	0	122	40

Table 4: Restricted Hessian spectrum summary at the embedded scalar extremiser (from `LT_B/result.json`).

across x, y , so the wording “one constant unitary” may be too strong. This concerns the form of an equality-case classification, not the inequality.

4 Comparator check of the scalar Lean formalisation

The repository `openai/math` contains a Lean 4 formalisation with a challenge file `ComparatorChallenges/LiebThirring.lean` (statement, ending in `sorry`) and a solution module `OAI.Analysis.LiebThirring.Main`. We installed upstream’s own pinned stack on a separate disk: Lean 4.34.1, Mathlib at upstream’s commit `d13f23b7`, and upstream’s `lakefile` (`autoImplicit` off), with no edits.

- `lake build` of the Lieb–Thirring closure (38 files) succeeds, and `#print axioms` for the four final theorems lists exactly `propext`, `Classical.choice`, `Quot.sound`.
- The `leanprover/comparator`, run with `landrun` (sandbox) and `lean4export` and the Lean default kernel, reports “*Your solution is okay!*” for `OAI.SharpLiebThirring.sharp_lieb_thirring` with permitted axioms `propext`, `Quot.sound`, `Classical.choice`.
- Deviations: `lean4export` and the Comparator were built from their tag `v4.34.0` sources with the toolchain file set to 4.34.1 (no 4.34.1 tag exists); `landrun` is commit `811cfff`; the sandboxed run needs the toolchain’s `bin` on `PATH` and `lake env`; no external kernel (nanoda) was run; the three controls were run after the main run.

γ	m	$\lambda_{\max}$	tolerance	$>\text{tol}$	within tol	$< -\text{tol}$	confirmed positive
3.0	1	5.23×10^{-1}	1.97×10^{-5}	9	0	9	5
3.0	2	5.23×10^{-1}	1.97×10^{-5}	27	18	27	5

Table 5: Negative control, $\gamma = 3$: positive directions are found, as they must be.

m	N	smallest excess over $\pi^2/4$	restarts within 10^{-5}
1	1	1.49×10^{-8}	9
1	2	2.31×10^{-6}	8
1	3	1.22×10^{-5}	0
1	4	1.27×10^{-4}	0
2	1	2.95×10^{-8}	9
2	2	5.08×10^{-7}	7
2	3	2.39×10^{-6}	4
2	4	1.55×10^{-5}	0
3	1	1.10×10^{-7}	6
3	2	1.89×10^{-7}	7
3	3	1.31×10^{-6}	6
3	4	4.17×10^{-6}	1

Table 6: Dual kinetic form (from `LT_C/result.json`).

- Controls for the Comparator, run on the same setup: a matching statement is accepted; a different statement is rejected (“theorem statement do not match”); a proof using an extra axiom is rejected (“Illegal axiom detected”).
- For the triangular-lattice energy theorem of the same repository we have only the native build under the same pin (229 files, no errors) and the axiom list of `OAI.AtomicTriangular.universal_energy_minimum`, again exactly the three standard axioms. A Comparator run on it was still in progress when this version was written, so *no Comparator verdict is claimed for it here*.

Process lessons. Before the pinned stack was available, an adapted copy compiled on a different Lean/Mathlib version needed a declared one-line rename for the Lieb–Thirring closure (`Set.equivOfEq` to `Equiv.setCongr`), while the triangular closure failed on a tactic whose behaviour changed (`convert/simp` left an unreduced `if`); both compiled unmodified under the pin. The lesson is that version drift is easy to mistake for a proof defect. An independent verifier of the first pilot also found that we had compiled with `autoImplicit` on while upstream’s lakefile turns it off; the final result above uses upstream’s setting.

What the Comparator does and does not certify. It certifies that the solution proves a theorem whose statement equals the challenge statement and uses only permitted axioms. It does not certify that the statement formalises the Lieb–Thirring conjecture (domain of test functions, extended-real sums, definition of the negative moment), which requires a human audit that has not been done. If the statement is faithful, it would settle the scalar sharp inequality for $1/2 < \gamma < 3/2$, which we understand to be an open conjecture in the literature (we did not search the literature exhaustively); this is precisely why the audit matters. It says nothing about the matrix-valued case.

5 Limitations

(i) The instrument resolves differences of about 10^{-6} only; a violation of relative size below that is invisible. (ii) The search is local and finite: 336 restarts at $m = 2$ cannot cover the space of potentials. (iii) The dual-form search did not converge in three cells. (iv) The near-maximiser analysis rests on a partial snapshot. (v) The Lean part is scalar. (vi) The statement audit is outstanding.

Reproducibility and acknowledgements

All numbers and tables in this paper are generated from the committed result files by `papers/lt_matrix_audit/gen_numbers.py`. Code, preregistrations, lane records with independent adversarial verification notes, and the Comparator logs, scripts and control sources are deposited in the accompanying bundle (and in the **AutoevolveAI** repository); the bundle's `MANIFEST.json` lists every file with its sha256. Large build outputs (oleans, checkpoints) are not included. The experiments, code and the drafting of this paper were carried out with the assistance of the AI model Claude (Anthropic) under the author's direction; the author is responsible for the content.

References

- [1] A. Laptev and T. Weidl, Sharp Lieb–Thirring inequalities in high dimensions, *Acta Math.* **184** (2000), 87–111.
- [2] E. H. Lieb and W. Thirring, Inequalities for the moments of the eigenvalues of the Schrödinger Hamiltonian and their relation to Sobolev inequalities, in *Studies in Mathematical Physics*, Princeton Univ. Press, 1976, 269–303.
- [3] `leanprover/comparator`, <https://github.com/leanprover/comparator>; `openai/math`, <https://github.com/openai/math> (both accessed 2026-10-08).